Airline Reservation System Problem Definition

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Problem and purpose

Customers need a clear way to find a dated departure, reserve seats for a small group, retrieve a confirmation and cancel without losing track of the reservation. Administrators need consistent airport, aircraft, route, schedule and fare records. Disconnected forms or spreadsheets can produce conflicting seat allocations, stale fares and incomplete passenger records. Airline Reservation System addresses these problems with a persistent MySQL reservation system and a reproducible analytics layer.

Users and workflow

A customer registers and logs in, searches a route and UTC departure date, reviews availability and fare, enters one to four fictional passengers, selects seats and confirms. The server calculates the total and saves a unique reference. The customer retrieves or cancels the booking. An administrator maintains the operational catalog, schedules departures, changes future selling fares and manages reservations. A protected dashboard summarizes booking outcomes, route occupancy and data quality.

Scope and constraints

The implementation uses Node.js, individual HTML pages, CSS and MySQL 8.4. A booking contains exactly one dated flight departure in one-way economy; it may contain one to four passengers. Seat selection is included in the displayed fare. Payment records are explicitly simulated. Layovers, hotels, cars, discounts, real ticket issuance, email delivery and real payment processing are excluded. The initial data includes four airports, two aircraft and 30 days of departures. Dates and times are labeled UTC. The system assumes a single classroom server and fictional personal information.

Measurable success criteria

A valid booking must return a unique 16-character reference and the correct server-calculated fare, persist after a server restart and expose only the owning customer’s records. Two simultaneous attempts for the same departure and seat must yield exactly one confirmed allocation. A failed multi-passenger booking must leave no partial records. Cancellation must release seats and void the simulated payment. Unauthorized administrative requests must be rejected. Three real users must complete search, booking, retrieval and cancellation; at least one also reviews administration. Automated test evidence supports technical criteria; real user sessions and a recorded presentation remain required before final submission.

Development assistance is recorded in the Codex log. The project owner must understand and explain the implementation. Deadline stated in the assignment: November 29, 2026.